SARAI CL-SEAMS DROUGHT MONITORING TOOL GUIDE

Using Standardized Precipitation and Temperature Indices

A. Overview & Objective

This document serves as the comprehensive Standard Operating Procedure (SOP) and technical reference guide for the SARAI CL-SEAMS Risk Mapping tool. It details the operational workflow for monitoring and analyzing agricultural drought and extreme heat conditions using the Standardized Precipitation Index (SPI) and Standardized Temperature Index (STI) within Google Earth Engine (GEE).

The primary objective is to evaluate spatial and temporal drought-heat risk, identify extreme compound climate events, and support agricultural impact assessment at the municipal level.

B. Frequently Asked Questions

What is SPI?

The Standardized Precipitation Index (SPI) is a widely accepted meteorological index used to quantify precipitation deficits or surpluses across multiple temporal scales. It is calculated by fitting historical precipitation data to a probability distribution (typically Gamma or Pearson Type III) and transforming it into a normal distribution with a mean of 0 and a variance of 1. Negative SPI values indicate dry conditions (below-average rainfall), while positive SPI values indicate wet conditions (above-average rainfall).

What is STI?

The Standardized Temperature Index (STI) measures thermal anomalies relative to long-term historical temperature baselines. Similar to the SPI, STI normalizes daily or monthly mean/maximum temperatures to standard deviation units. Positive STI values indicate hotter-than-normal conditions (heat stress), while negative STI values indicate cooler-than-normal conditions.

What are the Compounded Effects of SPI and STI?

When low precipitation (dryness) and high temperature (heat) co-occur, they create a Compound Drought and Heatwave event. The compounded impacts are non-linear and significantly more severe than individual extremes. It can trigger extreme crop heat stress, premature flowering/senescence, severe crop yield reduction, or complete crop failure.

Why is 3-month rolling SPI recommended for drought reports?

A 3-month SPI rolling window is recommended for operational agricultural drought monitoring because: (a) It aligns directly with typical seasonal crop growth cycles, (b) It provides an accurate reflection of medium-term soil moisture depletion in the root zone, and (c) It smoothes out short-term weekly precipitation variability while remaining sensitive enough to detect agricultural drought onset before critical hydrological impacts occur.

Table B.1. When to use other timescales.

Timescale	Primary Application	Impacts
1-month	Short-term moisture monitoring	Detection of sudden "flash droughts" and immediate crop germination stress.
3-month	Seasonal and agricultural monitoring	Root-zone soil moisture depletion, crop development stages, seasonal crop yield reporting.
6-month	Medium-term hydrological monitoring	Streamflow anomalies, river basin runoff, and reservoir level monitoring.
12-month	Long-term hydrological/environmental	Deep groundwater table depletion, long-term reservoir storage, and ecosystem structural impacts.

What are the classifications and threshold values of SPI and STI?

As standardized indices, the Standardized Precipitation Index (SPI) and Standardized Temperature Index (STI) measure the deviation of precipitation and temperature, respectively, from their established long-term climatological normals. Both indices generally produce values ranging from approximately −4 to +4, with 0 representing the baseline or normal condition. Values farther from 0 indicate increasingly greater deviations from normal conditions, with negative values generally representing drier or cooler conditions and positive values representing wetter or warmer conditions. The World Meteorological Organization (WMO) established standardized classification thresholds for these indices, as presented in the table 2 below.

Table B.2. Classification thresholds for SPI and STI based on standardized climate benchmarks (WMO).

Range of Values	SPI Classification	STI Classification
2.0 or higher	Extremely wet	Extremely hot
1.5 to 1.99	Severely wet	Severely hot
1.0 to 1.49	Moderately wet	Moderately hot
-0.99 to 0.99	Near Normal	Near Normal
-1.0 to -1.49	Moderately dry	Moderately cold
-1.5 to -1.99	Severely dry	Severely cold
-2 and lower	Extremely dry	Extremely cold

How does the SPI-STI persistence outlook differ with PAGASA drought forecaster?

The SPI-STI Bivariate Persistence Outlook differs fundamentally from PAGASA's drought forecasting approach in its scope, statistical methodology, and spatial precision. While PAGASA relies exclusively on a single variable—precipitation—categorizing conditions into Dry Condition, Dry Spell, or Drought based on fixed percentage reductions in rainfall over two to three consecutive months,

the SPI-STI tool measures compound climate risks by evaluating rainfall deficits (SPI) and heat stress (STI) simultaneously. Because both indices are standardized against a baseline, a specific index value represents the exact same statistical severity regardless of location, elevation, or season, whereas PAGASA's criteria depend strictly on historical local monthly averages.

Table B.3. PAGASA PH Meteorological Drought Monitor (*https://droughtcaster.pagasa.dost.gov.ph/)*

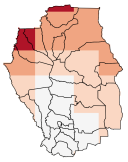
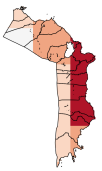
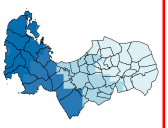
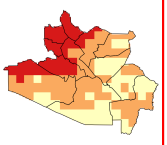
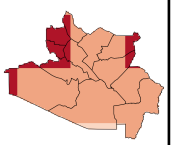
Normal	**Normal conditions.**
Dry Condition	Dry condition is defined as 2-consecutive months of below normal rainfall condition (21% - 60% reduction from average rainfall).
Dry Spell	Dry spell is defined as 3-consecutive months of below normal rainfall condition (21% - 60% reduction from average rainfall).
Drought	Drought is defined as 3-consecutive months of way below normal rainfall condition (>60% reduction from average rainfall).

How does the SPI-STI Tool and PAGASA Forecasts Complement Each Other?

The two systems operate on distinct forecasting horizons and spatial scales. PAGASA declares official drought classifications reactively once historical multi-month duration benchmarks have already been met on the ground. In contrast, the SPI-STI tool incorporates an AR(1) persistence carry-over model to project how current moisture and thermal stress will carry into near-future months, providing a forward-looking early warning. Powered by Google Earth Engine, it also delivers uniform spatial mapping at both provincial and high-resolution point levels, and it includes a thermal dimension that PAGASA's classification does not cover. This enables local decision-makers to identify specific municipalities facing acute compound hot-dry stress even when broad provincial forecasts report overall "Normal" status. This does not mean one system is inherently more accurate than the other; rather, it highlights the importance of utilizing available satellite-derived data to generate location-specific outlooks. Using the SPI-STI tool to complement official PAGASA forecasts allows agricultural planners and local leaders to make more proactive, well-informed decisions.

Table B.4. Comparison of provincial-level PAGASA Drought Classifications with SPI and STI index values.

Province	**PAGASA**				**SPI-3 (Precipitation)**	**STI-3 (Temperature)**
	July*	**August***	**September***	**October***	**October****	
Agusan del Norte						
Aurora						

Province	PAGASA				SPI-3 (Precipitation)	STI-3 (Temperature)
	July*	August*	September*	October*	October**	
Benguet						
Bukidnon						
Cebu						
Laguna						
Oriental Mindoro						
Pangasinan						
South Cotabato						

**July–September 2026 values reflect actual observed data, while the October 2026 outlook is forecasted based on the latest available data as of September 2026 (via https://droughtcaster.pagasa.dost.gov.ph)*

***SPI and STI outlooks computed using an AR(1) persistence model based on rolling August–September 2026 values*

Table 4 presents example comparisons between PAGASA's drought classifications and the SPI-3 outlook generated by the SEAMS tool. The PAGASA drought reports and SPI-3 results for Aurora, Benguet, Laguna, and Pangasinan are generally consistent for the month of October (highlighted in red). Both indicate an absence of drought impacts during the assessment period, generally reflecting normal to wetter-than-normal rainfall conditions. In Pangasinan, PAGASA reported normal conditions while SPI-3 indicated wetter-than-normal conditions in the western part of the province. This is because the PAGASA Drought Monitor classifies only below-normal rainfall

(dry condition, dry spell, and drought) and has no category for above-normal rainfall, so rainfall surpluses are reported simply as normal. This agreement in the data suggests that both approaches provide comparable assessments of precipitation conditions in these provinces.

However, in Agusan del Norte, PAGASA classified the province as experiencing a dry spell, while the SPI-3 results indicate near-normal precipitation across the majority of the province, with moderately to severely dry conditions in its northern portion. This likely reflects the spatial variability of rainfall within the province and the difference between a provincial classification and gridded standardized anomalies. Although the provincial assessment suggests relatively limited drought conditions overall, the SPI-3 map reveals localized areas experiencing more pronounced precipitation deficits, which highlights the importance of spatial analysis in identifying areas that may require more targeted agricultural monitoring and intervention.

In Cebu, South Cotabato, and parts of Bukidnon, PAGASA reported no drought for October, while the SPI-3 indicated moderate to severe dryness. In Bukidnon, the dry conditions are concentrated on the northwestern side, whereas Cebu and South Cotabato show widespread dryness, with some areas classified as extremely dry. It is important to note that PAGASA had classified Cebu and South Cotabato as experiencing dry conditions in September. Therefore, the dry anomalies indicated by SPI-3 in October may partly reflect precipitation deficits carried over from previous months within the three-month rolling window. Since SPI-3 accounts for accumulated precipitation over the preceding three months, it can capture persistent rainfall deficits that may not be apparent in assessments focused on the current month alone. These findings show the importance of considering the three-month timescale in agricultural drought monitoring, as it provides a broader perspective on seasonal rainfall conditions and potential crop water stress.

Lastly, in Oriental Mindoro, PAGASA classified the province as experiencing dry conditions, whereas the SPI-3 results indicate near-normal rainfall. This discrepancy may be attributed to differences in the assessment periods, rainfall aggregation methods, and spatial representation used by the two approaches. Near-normal SPI-3 values suggest that cumulative precipitation over the three-month period remains close to the historical baseline, even if rainfall during a more recent period has declined. Similarly, localized rainfall deficits may be obscured when precipitation is aggregated across a longer period or when provincial conditions are summarized. This emphasizes the importance of interpreting SPI-3 alongside PAGASA's drought monitoring reports and other available meteorological information to distinguish short-term rainfall deficits from persistent seasonal drought conditions.

The STI-3 adds a thermal dimension that is not part of PAGASA's rainfall-based classification, so it is best read as an additional layer that describes a different aspect of drought risk. In Cebu, South Cotabato, and northwestern Bukidnon, above-normal temperatures overlap with areas of SPI-3 dryness, indicating possible compound hot-dry conditions that may increase evaporative demand and crop water stress, even though PAGASA reported no drought for October. In Agusan del Norte, extremely hot conditions across the entire province occur alongside PAGASA's dry spell and the localized SPI-3 dryness in the north, which may further intensify moisture stress. In contrast, Benguet, Pangasinan, and Laguna show near-normal temperatures alongside near-normal to wet precipitation. Because STI-3 expresses temperature anomalies, not observed impacts, it is best interpreted as early-warning information to be considered together with PAGASA's official reports.

C. How to Use the CL-SEAMS Drought Monitoring Tool

Script URL:

https://code.earthengine.google.com/b361b3fcba524185d6f7485161cb0852

Dashboard Overview

The SPI-STI Bivariate Tool is a Google Earth Engine derived tool that shows where rainfall deficits and increased temperature occur together. It works at two levels: a point level analysis from a generated provincial bivariate map. The provincial boundary was further divided into respective municipal boundaries.

The tool turns rainfall into the Standardized Precipitation Index (SPI) and temperature into the Standardized Temperature Index (STI). Both indices are compared against the same 1990–2025 baseline for the same calendar month, a value of −1.5 means the same degree of unusualness in any place or season. It uses a uniform method in generating point level data and provincial level maps.

Table C.1. Code block section summaries

Section	What you do	What you get
1. Boundary Setup	Load a boundary asset and choose its name field.	Boundary outline on the map and a list of features (e.g. municipalities)
2. Point Level Analytics (optional)	Pick a feature or click the map, choose a timescale	SPI and STI time series, SPI–STI quadrant chart, 5-month persistence outlook
3. Generate Bivariate Map	Choose Observed or Persistence Outlook and a month	7×7 bivariate SPI-3/STI-3 map and a GeoTIFF download

Table C.2. Definition of Terms

Term	Meaning in this tool
SPI	How unusual a period's rainfall total is, in standard-normal units. Negative values correspond to drier than normal conditions.
STI	How unusual a period's mean temperature is, in standard deviations. Positive values mean hotter than normal conditions.
Timescale (k)	Length of the window in months: 1, 3, 6, 9 or 12 for points. We use a 3-month rolling timescale to generate the bivariate map.
Window end	The last month of a k-month window. A value labelled Aug 2026 at k = 3 covers Jun–Aug 2026.
Baseline	The baseline years started from 1990 to 2025. The years each month's statistics are fitted on.
Compound dry-hot	SPI ≤ −1 and STI ≥ +1 in the same window.
Outlook origin	The last observed window the persistence outlook starts from.
Lead	How many months after the origin the outlook window ends (1–5).
Persistence factor (φ)	The share of the current anomaly the outlook carries to a given lead. The values range between 0 and 1.

Table C.3. Data sources and reference periods

Variable	Earth Engine collection	Band and units	Update Frequency
Rainfall, final	UCSB-CHC/CHIRPS/V3/PENTAD	precipitation, mm per pentad (5-day period)	Updated monthly; 2-3 weeks lag time
Rainfall, preliminary	projects/climate-engine-pro/assets/ce-chirps-prelim-pentad	first band, mm per pentad	Updated weekly; 2-day lag time
Temperature	projects/climate-engine-pro/assets/ce-ag-era5-v2/daily	Temperature_Air_2m_Mean_24h, Kelvin (converted to °C)	Updated daily with 7-day lag time

The rainfall data are sourced from CHIRPS v3 and processed at its native 0.05° spatial resolution (approximately 5.5 km), while temperature data are obtained from ERA5. To avoid overlapping periods when combining final and preliminary CHIRPS v3 data, the tool identifies the date of the latest finalized pentad and appends only preliminary pentads dated after it. The six pentads for each month, with the sixth covering day 26 through the end of the month, are aggregated to calculate the monthly rainfall total.

Specifically, the script implements the following procedures:

- If preliminary pentads are unavailable after the latest finalized data, a warning appears in the status line below the title, indicating the number of missing pentads.
- If the preliminary data collection cannot be accessed, the tool uses finalized data only and displays a corresponding status message.

Table C.4. Reference periods

Period	Years	Used for
Baseline	1990–2025	Fitting the SPI gamma distribution and the STI mean and standard deviation, per calendar month, for points and maps alike
Point record	1990 to the latest complete month	The time series and quadrant charts (extraction starts in 1989 so 12-month windows ending in early 1990 are complete)
Persistence years	1991–2020	Estimating the persistence factors for the outlook (the current WMO climate normal period)

It is important to note that each rolling window is assigned to the year in which its final month occurs. For example, the December 2025–February 2026 window is attributed to 2026 and is therefore excluded from the baseline period.

Core Capabilities and Main Script Logic

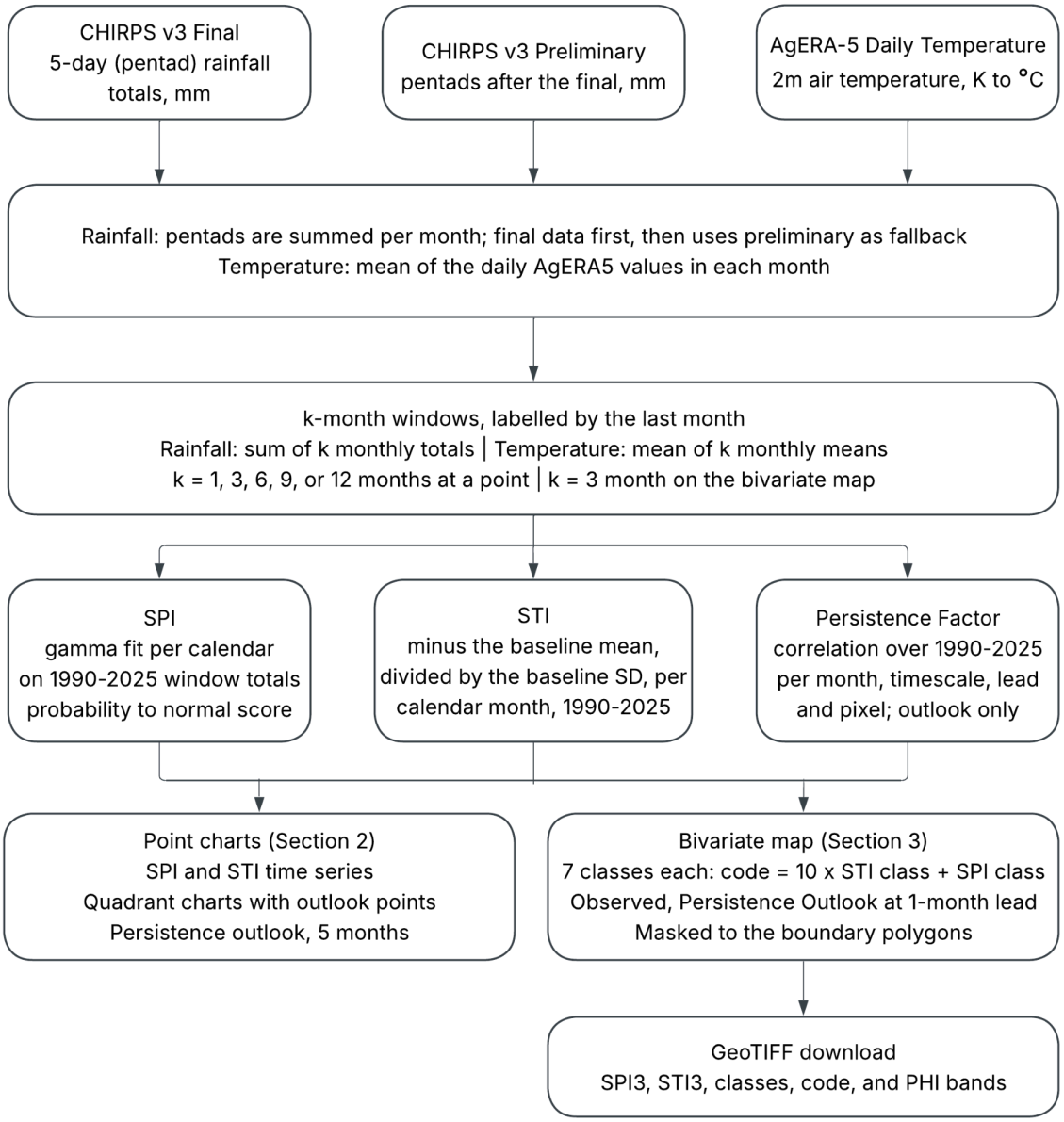


Figure C.1. Script logic and workflow

The figure above illustrates the tool's workflow, from data acquisition and processing to the generation of outputs. All chart values and map pixels undergo the same rolling-window calculations and SPI and STI fitting procedures, which ensures the consistency across outputs. For example, the point charts and bivariate map generator use the same calculation methods and pixel grid; therefore, at the 3-month timescale, the values displayed in a point chart should match those of the corresponding map pixel. The outlook is the only component that incorporates an additional persistence factor to project future conditions based on the continuation of existing trends.

The tool generates SPI and STI time series for any selected point from 1990 to the latest available month, using 1-, 3-, 6-, 9-, or 12-month timescales. It also provides a compound stress view that plots monthly SPI and STI values on an SPI–STI quadrant chart. This is used to highlight the

compound dry-hot conditions and identify where the persistence outlook lies in the quadrant. For spatial analysis, the tool generates a bivariate map of SPI-3 and STI-3 values for every pixel within the selected boundary. In addition, the persistence outlook projects current anomalies 1–5 months ahead for a selected point, based on the persistence of existing conditions. Finally, the tool allows users to download the map's index values, bivariate classes, and outlook factors as a single multi-band GeoTIFF file, directly to the browser without requiring Google Drive.

Table C.5. Script structure

Script section	Purpose
1. Configuration	Datasets, reference years, outlook settings, disclaimer text, colours
2. Bivariate palette	The 49 colours of the 7×7 matrix, codes 11–77
3. Dates and availability	Month arithmetic, last-complete-month check, joining final and preliminary CHIRPS
4. Math helpers	Gamma and normal functions, SPI and STI fitting, persistence factor (each with a point version and a map version)
5. State and UI	The left panel and its three sections
6. Overlays	The results panel and the bivariate legend
7. Boundary ingestion	Loading the asset, feature names and centroids
8. Point selection	Dropdown and map-click handling
9. Point extraction	One request that returns the whole monthly series at the point
10. Point statistics and charts	SPI, STI, outlook and the three charts
11. Bivariate raster engine	Map indices, outlook factors, classification, Section 3 controls
12. Download	GeoTIFF link with automatic retries
13. Startup	Availability check and default month

Sources

- CHIRPS v3 pentad, Earth Engine Data Catalog
- CHIRPS-Prelim, Awesome GEE Community Catalog
- CHIRPS v3, Climate Hazards Center
- https://gee-community-catalog.org/projects/agera5_datasets/
- ee.Image.gammainc, Earth Engine API reference
- WMO SPI User Guide (WMO-No. 1090, 2012), for SPI categories and the gamma method